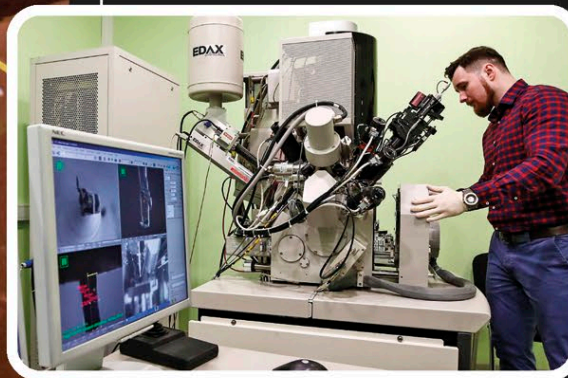


ZOOMING IN

MICROSCOPES

Microscopes have been used by humans for hundreds of years to study the smallest parts of our world—from the cells in our own bodies to tiny microorganisms smaller than a grain of sand. Different microscopes can show different levels of detail. Light microscopes use lenses and can magnify things 1,500 times, while more powerful electron microscopes use beams of tiny particles called electrons, which can help magnify objects up to a million times their original size.

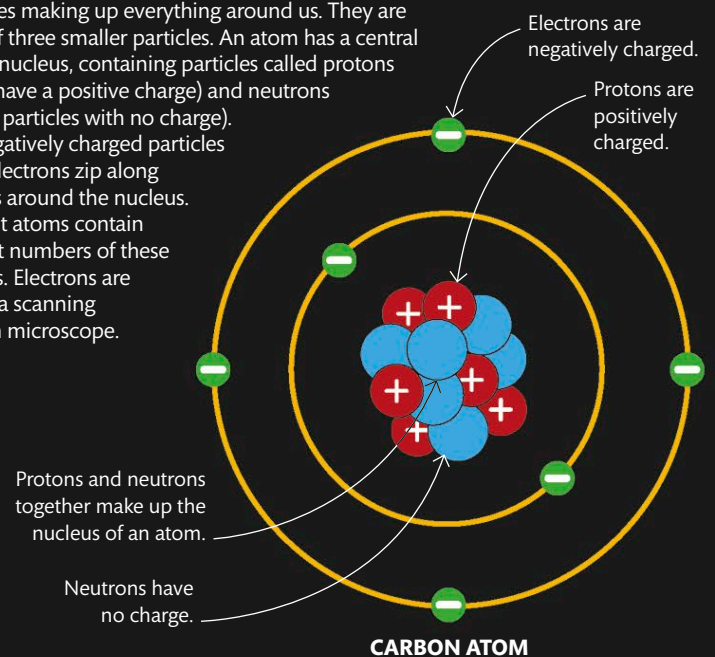


SEM

At a research laboratory in Russia, this scientist is using a type of microscope called a scanning electron microscope (SEM) to analyze fragments of an object. SEMs can be used to study bacteria, to tell apart rock samples, and even to help assemble tiny parts of computers.

ELECTRONS

Atoms are the building blocks of the Universe—tiny structures making up everything around us. They are made of three smaller particles. An atom has a central hub, or nucleus, containing particles called protons (which have a positive charge) and neutrons (neutral particles with no charge). Tiny negatively charged particles called electrons zip along in orbits around the nucleus. Different atoms contain different numbers of these particles. Electrons are used in a scanning electron microscope.



4 Close-up creatures

Under the high magnification of an electron microscope, the details of a dust mite—only about a quarter of a millimeter long—can be seen. Scientists can learn about the creature's anatomy from this close-up view.